Vortex86MX+ (A9130)
Layout Guideline

Ver. 0.3

Modify	Date (mm/dd/yy)	Version	Description
Joe Yen	01/19/11	0.1	Initial Release
Joe Yen	10/30/2012	0.2	1. Modify DDR2 Layout guide for 1G bytes system 2. Add power on/off sequence 3. Modify the ball pin for the TX VSS of Ethernet. 4. Modify the stack up for 6-layer and 4-layer PCB
Joe Yen	12/14/2012	0.3	1. Add the crystal recommend for 14.31818MHz and 32.768KHz. 2. Add RXC pull down reference circuit. 3. Add the delay time from 1.8V to 3.3V for power on sequence 4. Add the reference circuit of level shift for VGA port.

Overview

1.1. General Description

The following document provides some technical data that describe layout guideline of A9130. The main purpose of this document was to provide A9130 with high quality signal for system reliable operation.

1.2. PCI bus Layout Guideline

Control the skew of PCI clock between A9130 and PCI device within 2ns. Routing signals of PCI with 5mil trace width and keep minimum 10mil space. Add a 100ohm resistor on each IDEVSEL input of PCI devices. Route PCI trace length within 8 inch. If application need to combine PCI slot and down devices. Place down devices on PCI bus end to improve signal integrity. A9130's PCI interface isn't support 5V tolerance. Please don't connect to PCI device of 5V. Some signals need to pull up or pull down for the interface that we recommend using 10k ohm resistors.

1.3. IDE bus Layout Guideline

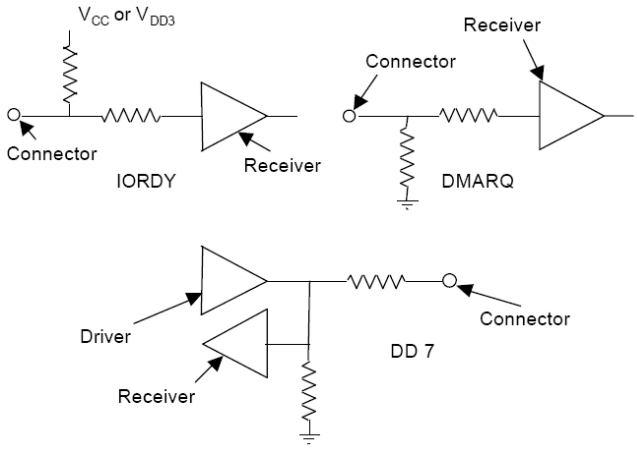
Matches trace length of IOR- and IORDY within 100mil. Matches trace length of IOR-/IORDY and D0~D15 within 400mil. Control the trace length of IDE interface within 6 inch. Routing signals of IDE with 5mil trace width and keep minimum 10mil space. We recommend to buffer DASP- signal to driver LED to reduce EMI issue. The pull up and pull down is required for interface on host side as follows:

1. The DMARQ needs a 5.6k ohm pull down resistor.
2. The IORDY needs a 4.7k ohm pull up resistor.
3. The INTRQ needs a 10k ohm pull down resistor.
4. The CSEL connect to ground.
5. DASP- needs a 100k ohm pull up resistor if buffer the signal to driver LED.

The series termination required for the interface on host side as follows:

1. The IOR-, IOW- and DMACK- need a 22ohm series termination resistor.
2. The CS0-, CS1-, DA0~2, D0~15 and RESET- need a 33ohm series termination resistor.
3. The DMARQ, INTRQ and IORDY need an 82ohm series termination resistor.

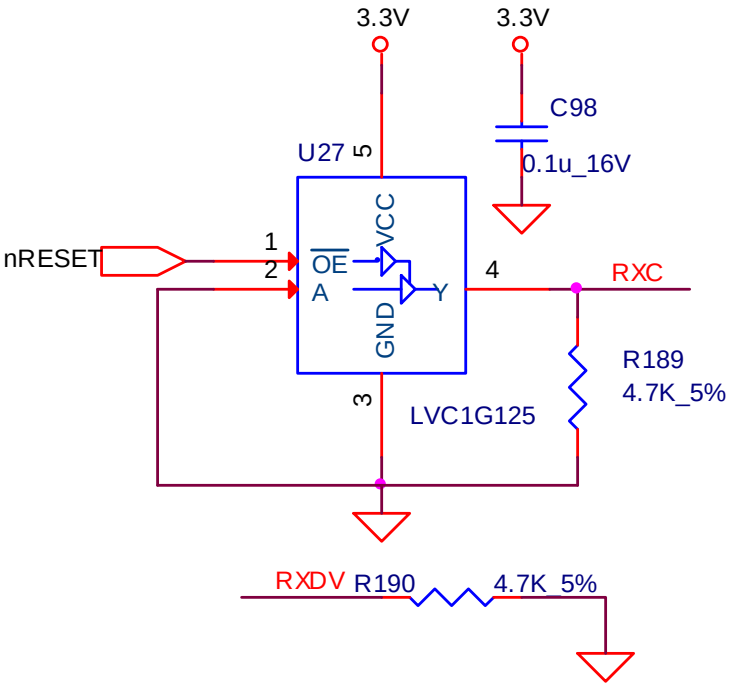
The topology for series termination with pull up or pull down is as follows:



Place all series termination near IDE connector.

1.4. Ethernet PHY Layout Guideline

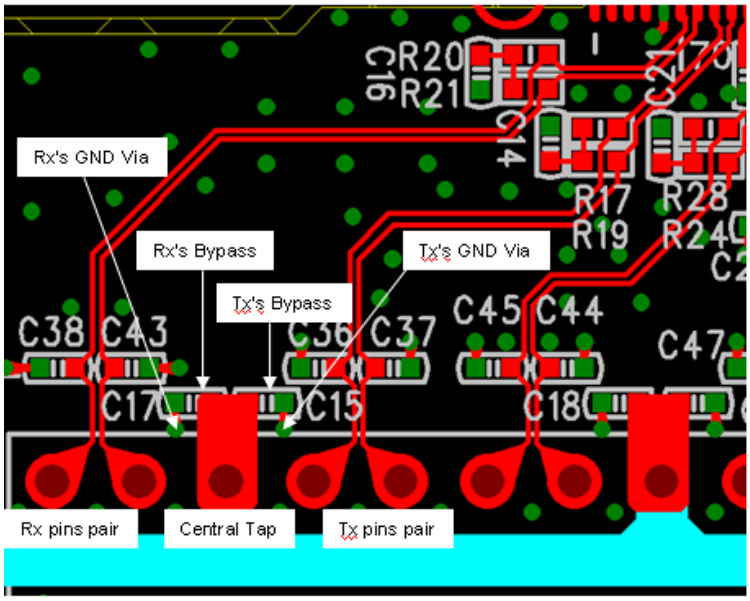
RXC and RXDV must be pull down for normal operation .The recommend reference circuit is as follows for the pull down scheme.



Please supply all of Ethernet powers and to power down the internal Ethernet by register if the internal Ethernet no need. The location of termination resistor is importance for Ethernet. Place termination resistor near IC side. Route differential pair of Ethernet PHY with 4.5mil traces width and 5mil space between TXP (RXP) and TXN (RXN), keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place resistor of ISET near A9130 at least with trace width of 20mil. Place Transform near RJ45 connector and place termination resistor near IC side.

Pin B15 (AVSS1_EPHYTX33) is the ground pin of transmitter; it can't share via with other analog ground pins. It

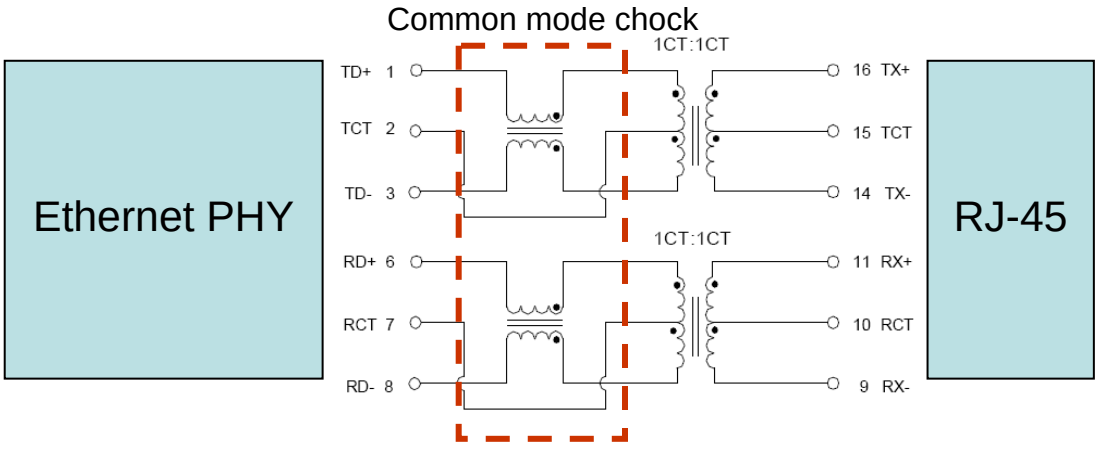
can't through bead connect to system ground. Strong suggestion connection to clear system ground directly. It is importance for the layout of transformer's central tap. Due to our PHY support auto MDI/MDIX. Add the bypass capacitor of central tap for transmit path and receiver path with ground via near transmit and receiver pins of transformer respectively. The figure is as follows:



Ethernet's Layout guideline

The LAN transformer selection guide is as follow:

Select the LAN transformer with common chock on the PHY side to enhance EMI's performance.

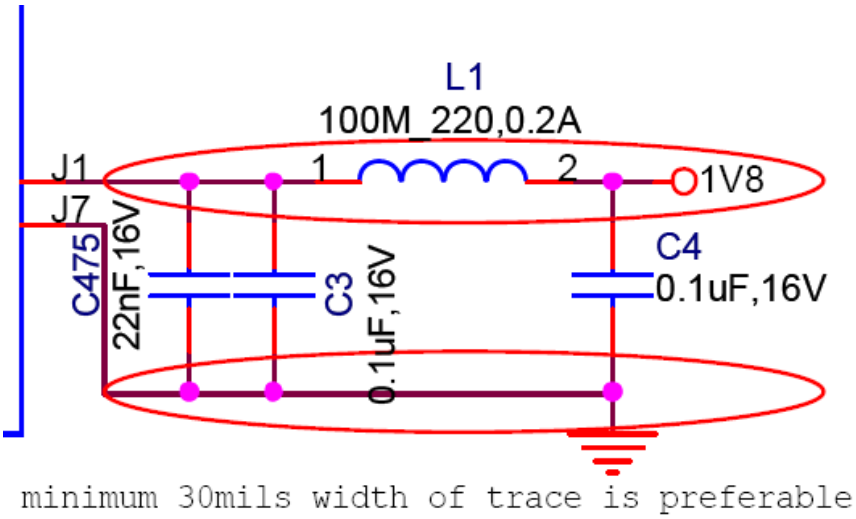


1.5. USB PHY Layout Guideline

Route differential pair of USB PHY with 5mil traces width and 5mil space between DP and DN, keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place the resistor of ISET near A9130 at least with trace width of 20mil. Place common chock near connector if present.

1.6. PLL layout Guideline

A clear power supply is importance for PLL in high performance CPU. DMP recommend a filter technology to reduce jitter as follows:



1.7. SD Card Interface Layout Guideline

Route traces length of SD Card Interface within 1000mil.

1.8. 1G bytes DDR2 Layout Guideline

According to the below-mentioned guideline, some constraints were appended as follows:

1. Control the impedance of trace between 55 and 60 ohm.
2. Control the trace length of DQS/DQS# between 1700 and 2000 mil.
3. Control the trace length of DQ/DQM is shorter than DQS/DQS# between 0 and 700 mil.
4. The damping resistor is 10 ohm for data bus.
5. The damping resistor is 0 ohm for address, command and clock bus.
6. The termination resistor is 100 ohm for clock bus and the location is at the first branch for A9125 to DDR2.
7. Control the trace length of CLKP/CLKN is longer than DQS/DQS# between 0 and 300 mil.

1.9. DDR2 Interface layout Guideline

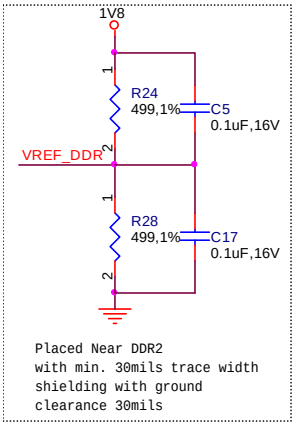
More attention must be paid to high speed rule for board layout and bus loading to make sure that short clock cycle time can be met without noise, ringing, crosstalk power or ground bounce. To provide a good return path and limit above noise on the DDR interface, all system memory signals should reference a single contiguous plane as possible.

The recommendation layout guideline for a DDR device is as follows:

1. In order to attain highest speed as possible. We suggest the signals of DQS/DQS#, DQM and DQ only a loading preference. Please connect with DMP if the signals of DQS/DQS#, DQM and DQ need to load more than one.
2. The DDR clock is differential signal that need a 100 ohm differential terminal resistor near DDR side for a DDR configuration or on split point for two DDRs configuration, and keep the trace length under 0.2 inch from split point to respective DDR. Reserve location of two 33ohm damping resistors near CPU side for low power application. The two trace width is 4.5 mil and space 5mil to get near 100ohm differential impedance, and keep minimum 15 mils space for other traces to reduce crosstalk as possible. The trace length of two differential signals must be match within 10mil. The category signals were routed within 1.4 inch.
3. Routing each signals of byte lane on same layer with same via count. Control the number of via within 2. The mismatch of trace length must be within 20mil for each signals of byte lane. The high byte lane is DQS1/DQS1#, DQM1 and DQ8~DQ9 and the low byte lane is DQS0/DQS0#, DQM0 and DQ0~DQ7 respective. Place a damping resistor on middle of each signal to trade off the timing of read and write for low power with ODT off. The DQS/DQS# is differential signal that they were routed differential with 4.5mil trace width and 5mil space between DQS and DQS#, keep minimum 15 mils space for other traces to reduce crosstalk as possible. The other signals were routed 5mil trace width and keep minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The high byte lane and low byte lane must be matching within 0.2 inch. The category signals were routed within 1.2 inch. It is importance to match the differential signals of DQS and DQS# within 10mil.
4. Routing signal of command and address with a 33ohm damping resistor near CPU side. The technology of VTT terminal was prohibited since the address pins of DMP support bootstrap function. The category signals were routed 5mil trace width with minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The signals of command and address must be matching within 0.1 inch. The category signals were routed within 1.5 inch.
5. To provide a good return path, all system memory signals should reference a single contiguous plane as possible.
6. DDR memory signal trace edges must be at least 4 times (20 mils) trace width from any non-DDR memory signal trace edge.
7. Routing trace near edge of the reference plane should keep minimum 5 times trace width from edge of signal trace to edge of the reference plane.
8. The signals of address and command must be following Balanced-T routing rule for the application of two DDR devices.
9. The generation of Vref is critical. It must be follow the variance of VDDQ. We are suggestion to generate the reference from VDDQ by two 499ohm divide resistors with parallel 0.1uf X7R type bypass capacitor for each

resistor. Place the divide resistors and bypass capacitor on quiet area. Also place a 0.1uf X7R type bypass capacitor on Vref pin of each device.

10. The circuit of Vref is as follows:



11. Routing Vref as short as possible with minimum 30mil trace width and at least 30mil space from other signal.

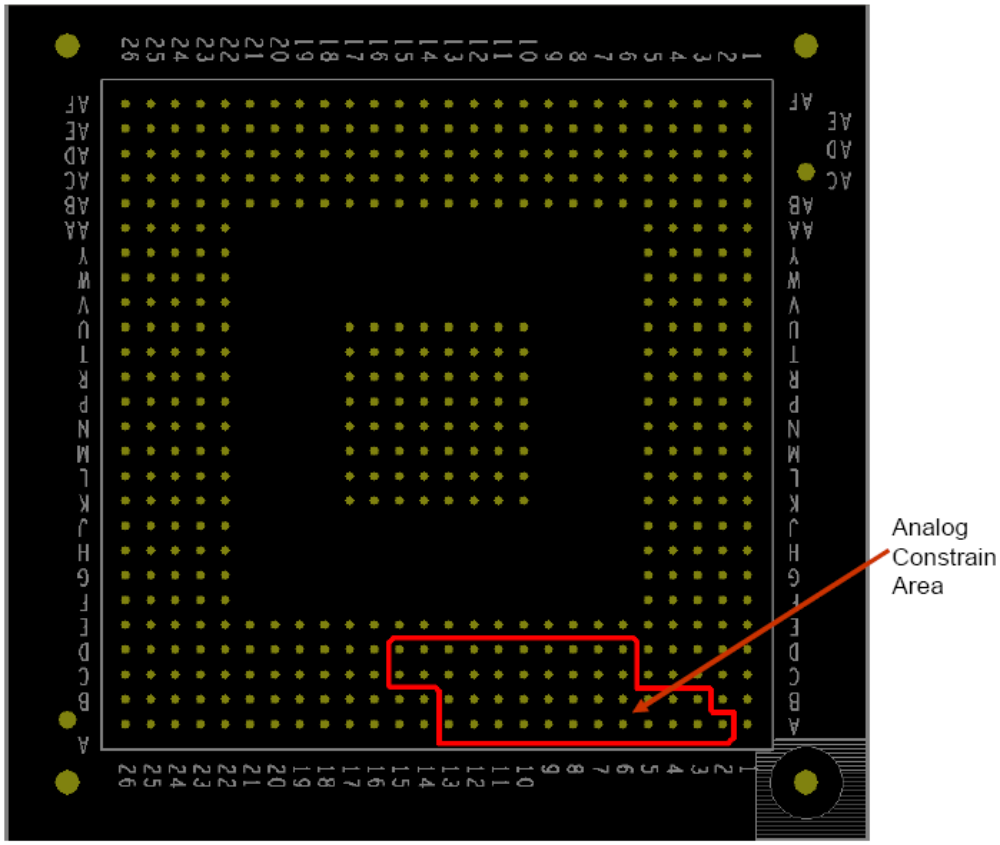
12. Matching trace length by serpentine line of “S” pattern. Make sure at least 5 times the distance of the trace to reference layer between line segments of “S” that parallel for the application of microstrip.

13. We are suggestion placed a bypass capacitor near vias every 4 to 5 vias for signal transition reference plane.

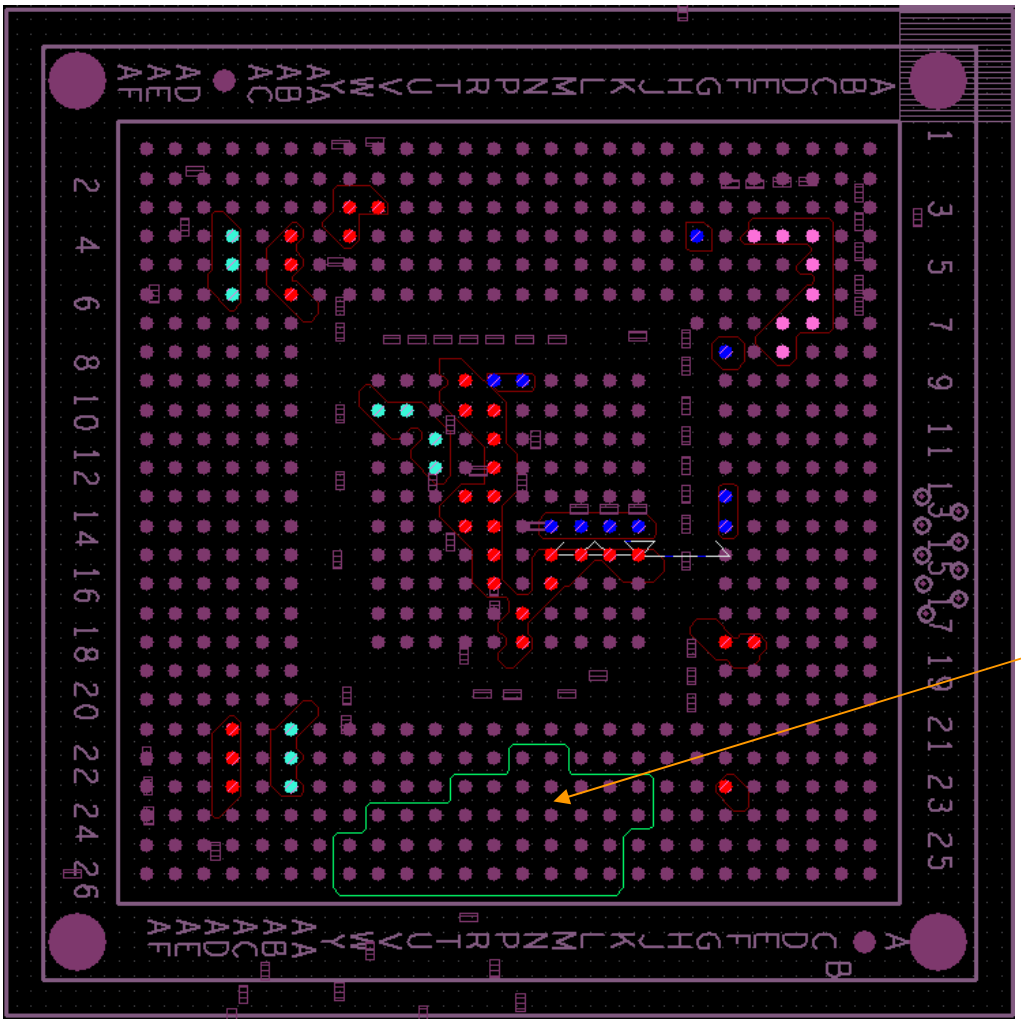
14. Place the resistors of bootstrap within 0.2 inch the pins of DDR2 device.

1.10. Analog area Layout Guideline

Avoid any digital signal cross the area. The constraint area for A9130 is as follows:

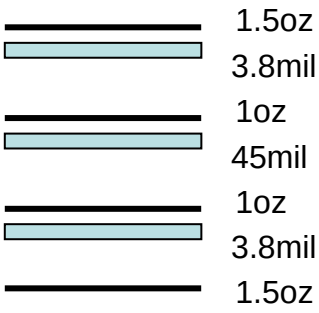


The constraint area for A9130 is as follows:

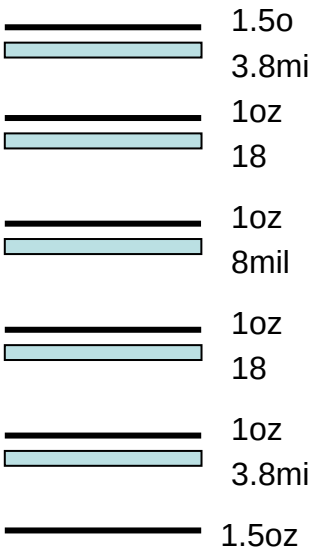


1.11. PCB Stack up

The recommendation PCB stack up is as follows for 4 layers PCB:



The recommendation PCB stack up is as follows for 6 layers PCB:

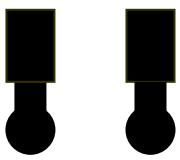


1.12. Bypass Capacitor scheme

The role of the bypass capacitor is to decouple the power supply bus from the IC. Bypass a power bus from the device internal noise requires capacitor with very low inductance. The objective is to eliminate the effects of the power bus inductance and resistance so that transient currents flowing across the power bus do not cause excessive noise at the power and ground pins of the IC. Therefore, the bypass capacitor should have low effective series resistance (ESR) and series inductance (ESL) while having a large enough capacitance value to supply current to the IC during switching.

The recommendation bypass scheme for the large surge current is as follows:

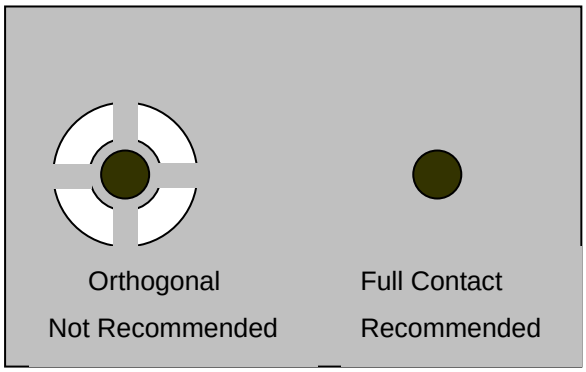
1. It is recommended that at least a via per power pin for SDRAM memory interface
2. It is recommended that multiple vias per bypass capacitor to further reduce inductance.
3. It is recommended that X7R or X5R type and 0402 size capacitor for bypass.
4. It is recommended that a 47uf bulk capacitor for the interface.
5. The land pattern of bypass capacitor is as follows:



1.13. Suggestion Via for DDR2 and A9130

In order to minimum the series inductance, ground and power via should be maximized as possible. We are

suggestion on DDR deice that via is at least 18 mil pads with 10mil drilling size. We are also suggestion on A9130 device that via is at least 20 mil pads with 10mil drilling size. The via pattern connect to power/ground plan is as follows:



1.14. POWER UP SEQUENCE

The Power on sequence is 1.0V 1st, 1.8V 2nd and 3.3V last. The delay of 1.8 V to 3.3 is more than 5ms recommendation.

1.15. POWER OFF SEQUENCE

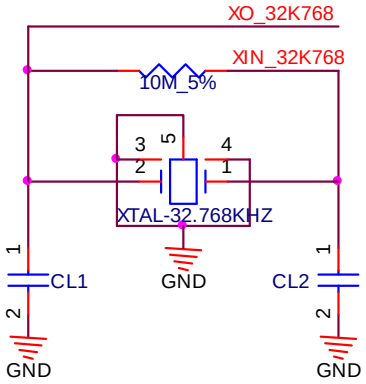
The Power off sequence is in the reverse order as power up

1.16. Crystal recommend TBD for 32.768KHz and 14.31818MHz

The recommend specification of 32.768KHz crystal is as follows:

1. The product must meet RoHS Directives.
2. Aurum Part Number : XF66M400003290
3. Holder Type : FORK-2060
4. Center Frequency : 32.768 KHz
5. Frequency Tolerance (at 25+/-3 deg.C) : +/- 20 ppm max.
6. Frequency Stability : -0.036+/-0.006 ppm/deg.C^2
(over operating temperature)
7. Operating Temperature : -10 ~ 70 deg.C
8. Storage Temperature : -40 ~ 85 deg.C
9. Equivalent resistance(ESR) : 35k Ohm max.
10. Load Capacitance (CL) : 9.0 +/-0.1 pF
11. Shunt capacitance (C0) : 1.7 pF typ.
12. Capacitance ration : 550 typ.
13. Drive Level (DL) : 1.0 uW typ.
14. Insulation resistance : 500M ohm min. at D.C 100V

The recommend circuit of 32.768KHz crystal is as follows:



The parasitical capacitance of 5mil PCB trace width on outer layer is about 1.1835pF/cm for 3.8mil P.P reference to shield layer.

The total capacitance of CL1+x*1.1835+3=CL1_total

Where x is trace length crystal pin to Xout pin of A9130.

The total capacitance of CL2+y*1.1835+3=CL2_total

Where y is trace length crystal pin to Xin pin of A9130.

CL=CL1_total*CL2_total/(CL1_total+CL2_total)

CL is the load capacitance of crystal.

The recommend specification of 14.31818MHz crystal is as follows:

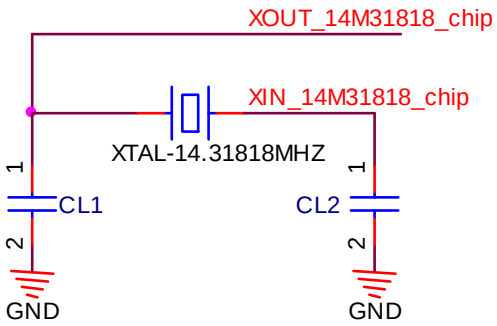
PACKAGE : L50 (5.0 x 3.2 x 1.3) mm

ITTI PART NO. : 1300712-14.31818MHz

ELECTRICAL CHARACTERISTICS :

Parameter	Symbol	Condition	Specification	Unit
1. Nominal Frequency	Fo	At 25°C	14.31818	MHz
2. Oscillation Mode	Fundamental			
3. Frequency Tolerance	ΔF/Fo	At 25°C	± 20	ppm
4. Tolerance Stability Over Temp. Range	Tc		± 20	ppm
5. Operating Temp. Range	Topr		-20 to +80	°C
6. Storage Temp. Range	Tstg		-40 to +85	°C
7. Load Capacitance	CL		12	pF
8. Equivalent Series Resistance	R1		60 max.	Ω
9. Drive Level	DL		300 max.	μW
10. Aging	Fa	At 25°C, Per Year	± 5 max.	ppm

The recommend circuit of 14.31818MHz crystal is as follows:



The total capacitance of CL1+x*1.1835+3=CL1_total

Where x is trace length crystal pin to Xout pin of A9130

The total capacitance of CL2+y*1.1835+3=CL2_total

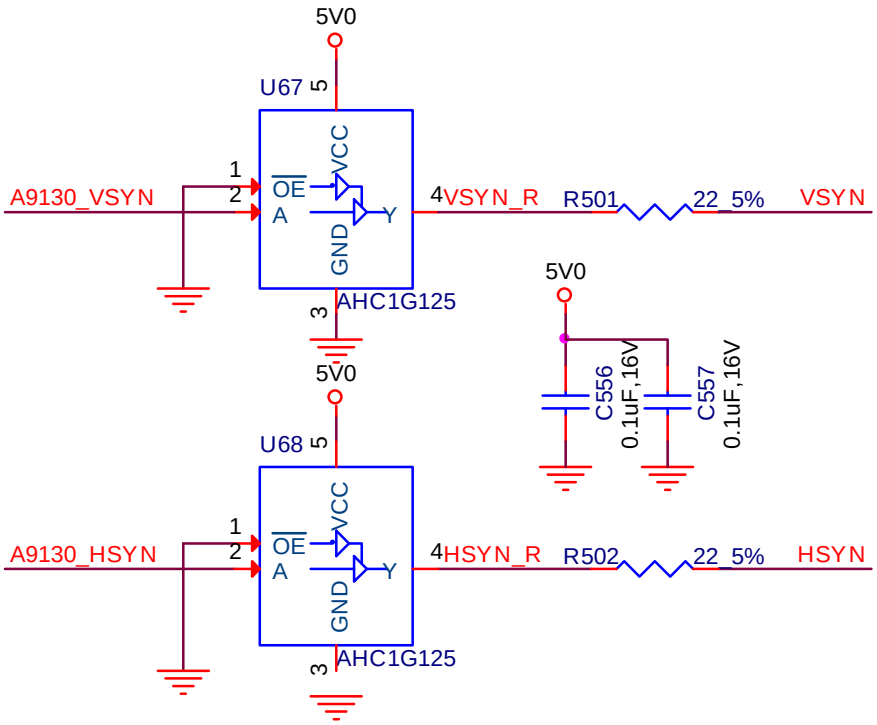
Where y is trace length crystal pin to Xin pin of A9130.

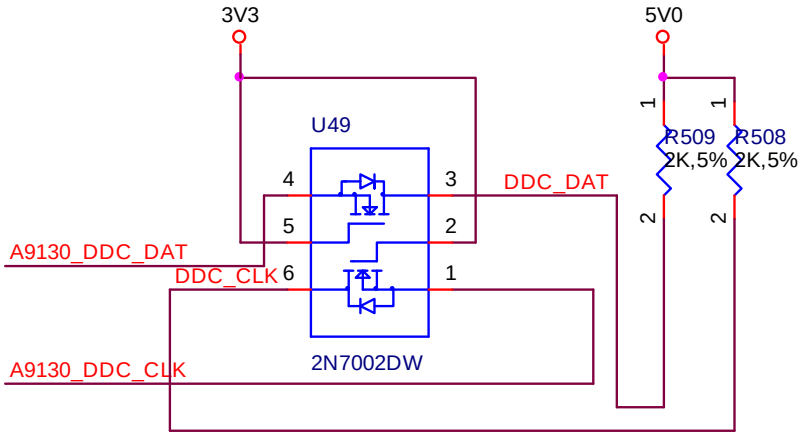
CL=CL1_total*CL2_total/(CL1_total+CL2_total)

CL is the load capacitance of crystal.

1.17. The reference circuit for VSyn and Hsyn of VGA

To drive traditional CRT monitor by our VGA port. We recommend adding level shift on the output of Hsyn, Vsyn and DDC channel.




3V3
5V0
U49
4
3
5
2
6
1
DDC_DAT
A9130_DDC_DAT
DDC_CLK
A9130_DDC_CLK
2N7002DW
R509
2K,5%
R508
2K,5%
1
1
2
2